A KNOW HOW THEORY OF TECHNOLOGY

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Abstract. This paper develops a formal theory of organizational capabilities by breaking production up into a series of small steps called activities. Activities must be chained together in a routine in order to produce an output. Thus a routine also formalises the way in which output depends on a firm's capabilities. The theory provides a new foundation for neoclassical production theory in which synergies and new products can be generated by firms merging their capabilities. These new aggregation results deviate from the Neoclassical Theory of the Firm and have important implications for industrial policy and anti-trust regulation.

1. Introduction

The Theory of the Firm, as advanced by Coase, Williamson and Hart describes production in which multiple steps in the production process are located inside a firm and transacting the associated intermediate products is economically unattractive and as a result no market for these specialized intermediate products operates. While a great deal has been written as to why this situation might occur the consequences for the Neoclassical theory of production have been overlooked.

The transaction cost and property rights approaches describe, quite realistically, firms in which the technology consists of a number of small steps. Neoclassical production theory has always treated these small steps as a black box and focuses instead on the end-end relationship between inputs and outputs. We will show in this paper that considering these small components of technology and production give rise not just to a descriptively more accurate view of how a firm operates but new results on how the technology and productive capabilities of a firm are changed by the acquisition of new production level know how.

1.1. **Example.** Benefits of merger/acquisition illustrated by graph.

Key words and phrases. Routine, Capability, Grammar, Organization.

The first author is grateful to the second author.

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Nodes in the graph are states in the production process. The firm knows how to transform some states into other states (as indicated by the edges). We refer to these small transformations as *activities*.

As shown in Figure 1, each firm has its own set of activities and staring at the empty state, denoted λ is able in three steps to produce state σ_4 which happens to be a saleable final good. While each firm produces the same output, and does so using the same number of activities, i.e. in three steps, the production processes are different.

Firm 1 uses, in order activities a_1, a_2, a_3 and passes through the states $\lambda, \sigma_1, \sigma_2, \sigma_4$. While Firm 2 uses activities a_4, a_5, a_6 and passes through states $\lambda, \sigma_2, \sigma_3, \sigma_4$.[1]

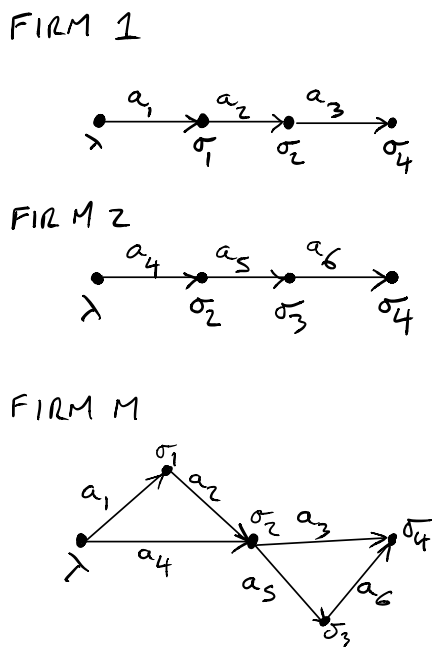

FIGURE 1. Simple technology graphs

Let us now consider what happens to the technological possibilities under a merger or acquisition. Let's call the merged firm M, and we assume, as is standard, that its technology is the union of all things that the two original firms knew. In the neoclassical setting this would simply mean that M had two different of producing σ_4. In our setting somethings else is possible.

[1]Each production has an input requirement set, which when combined with factor prices would give a minimum cost of producing that transformation (which in general will be non-unitary).

The technology of firm M is described at the bottom of Figure 1 by the graph made from the union of the two original graphs. Producing σ_4 involves finding a path from λ to σ_4 in the graph for Firm M.

For Firm M it is now also possible to produce σ_4 from λ by two additional sequences of activities: $a_1a_2a_5a_6$ and a_4a_3. Assume the total cost was the same for Firm 1 and Firm 2, then unless the cost of a_4 is the same as the cost of a_1a_2 and the cost of a_3 is the same as the cost of a_5a_6 one of the new sequences of activities will be strictly cheaper than the activities used by firm's 1 and 2.

For convenience we shall assume in this example that the technology of each firm is unitary, by which we mean the cost of each production is normalised to \$1. Thus each firm is able to produce the state σ_4, say one unit of a final good, at cost of 3.

By inspection the shortest such path uses

2. Definitions

2.1. **Basic definitions.** A key idea is that a very complex technology can be decomposed into a (relatively) small number of (possibly repeated) simple activities.

A technology describes the ability of an organization or person to achieve a certain action or outcome. More specifically a technology describes all of the transformations an organization knows and is able to perform between various states. This idea is made more precise below. Operationally the critical point is that all the activities taking place within the organization are built up from the components of the technology, including the way particular technologies are implemented within the organization, communication, etc.

We describe an organizational technology via the following formalizations.

Definition 1. Let Σ be a finite set of *descriptors*. A *production state*, or just *state*, is a finite string of descriptors:

$$s = \sigma_1\sigma_2\cdots\sigma_n,$$

where $\sigma_i \in \Sigma$ for $i = 1, 2, \ldots, n$.

A descriptor might be something as simple as "blue" or "square", so that a production state describes a complete or incomplete product. However for other applications the relevant descriptors might be more complex, for example finished goods like SUV, sports car or hatchback. In this case a production state refers to a sequence of final products.

Definition 2. Σ^* is the set of all production states generated by descriptors in Σ. It is convenient to also include the null state λ, that has no descriptors in it.

With regard to production, Σ^* describes all conceivable states of the world, although they may not of course all be feasible with a given technology. We exclude from consideration at this point time, location and exogenous shocks from the definition of descriptors and hence production states. Just as in standard neoclassical production theory, these issues can be addressed by indexing the descriptors appropriately.

2.1.1. *Example (Tom Sawyer).* For example, consider the production of painting a wall. In our framework this action has a simple representation as a single production. In the neoclassical framework this action would be described by a perhaps infinite number of ordered pairs with the first element of the pair being each possible unpainted wall, and the second element the corresponding painted wall.

2.1.2. *More definitions.* The saleable products in our framework is the subset of production states for which there is a market for the production state, which gives the following definition:

Definition 3. The *production set* is $\Omega \subset \Sigma^*$. We refer to an element $\omega \in \Omega$ as a *product*.

Our approach to production is fundamentally discrete. Whether a production state describes a sequence of individual goods or the characteristics of a single good the quantities involved are of finite arbitrary precision rather than real numbers. This is realistic since most, if not all goods, are available in finite precision quantities, which is to say there is some smallest unit of any relevant feature which is measurable and transactable.

The great advantage of our discrete production approach is the simplicity with which it allows us to describe the characteristics of differentiated products. In our modern increasingly technologically sophisticated world the ability of a production theory to describe new and differentiated products, rather than the quantities of a finite list of commodities, is critical. In our framework even a small number of descriptors can generate a countably infinite number of (finite length) production states.

In many cases our descriptors will be the familiar product characteristics from the theory differentiated product models of industrial organisation. However since the theory developed here is of production not consumption we use the term *descriptor* to emphasize that production might involve intermediate steps which are not relevant to consumers, or which are indistinguishable to consumers but require different resources or processes in production.

Two additional features of the discrete production framework deserve noting. First of the set of descriptors is not necessarily ordered so no

notion of cardinality amongst descriptors is imposed (although this can be added as an additional assumption in a particular application). Second, the descriptors in a production state do have an ordering so that sequencing is an inherent part of our approach. Neither of these features is readily accommodated by standard, finite neoclassical production theory and would require shifting to infinite dimensional commodities. While it is probably possible to replicate our approach using infinite dimensional commodities history has indicated that such an approach lacks tractability in dealing with differentiated products and the finer details of production.

The final component of our theory is the notion of an *activity*, which describes something an organization/firm is currently capable of doing in terms of the production process.

Definition 4. An *activity* a is a rewriting rule on production states which transforms one substring into another substring.[2] Let s and s' be substrings, $s, s' \in \Sigma^*$, we will write an *activity* as a which rewrites s as s' as $a : s \to s'$ or alternatively

$$s \xrightarrow{a} s',$$

by which we mean, for all $s_1, s_2 \in \Sigma^*$

$$a : s_1 s s_2 \to s_1 s' s_2.$$

The set $A = \{a_1, ..., a_m\}$ is the finite set of activities a_i available to a firm.

Implicitly a technology is defined for activities operating on a given set of descriptors and empty/start state λ.

The important thing about activities is that they rewrite, or transform, specific substrings not specific production states. The same string may be a substring of many states and hence one production can potentially transform many states. This is a way of describing the fact that a specific technique may be used in many different situations.

Definition 5. Given the set of activities A, a *routine* ρ is a string of not necessarily distinct activities $a_1, a_2, \ldots, a_n$ that transforms a state $s \in \Sigma^*$ into some other state $s' \in \Sigma^*$.

For example we might have $\rho = a_1 a_2 \ldots, a_n$ with

$$s \xrightarrow{a_1} x_1 \xrightarrow{a_2} x_2 \ldots \xrightarrow{a_n} s'.$$

More concisely it is often convenient to write $s \xrightarrow{\rho} s'$.

[2]In Computer Science string rewriting rules of this form are known as semi-Thue processes, see **?**. The connection we make with economic production is however novel.

By the length of a routine we mean the length of the string of activities which define the routine. Similarly, the$i$th element of the string of activities defining a routine ρ is denoted $\rho[i]$.

Remark: To avoid ambiguity we will follow *left derivation* in constructing routines.[3]

A set of activities A may permit multiple ways in which one state s can be transformed into another state s'.

Definition 6. Given a set of activities A the set of routines that transform s into s' is $\rho(s, s')$. For convenience we write $\rho(\omega) \equiv \rho(\lambda, \omega)$ for $\omega \in \Omega$.

Definition 7. A *technology*, denote by $\mathcal{R}(A)$, is the set of all routines generated by the set of activities A. That is, for each $\rho \in \mathcal{R}(A)$ there exist $s, s' \in S$ such that $s \xrightarrow{\rho} s'$

Definition 8. The *output set generated by a technology* $\mathcal{R}(A)$ is the set $\Omega(A)$ of all the outputs ω in Ω that can be produced by a routine ρ in $\mathcal{R}(A)$ starting from the null state λ:

$$\lambda \xrightarrow{\rho} \omega.$$

Some natural relationships arise immediately from these definitions. If P, Q are sets of activities such that $P \subset Q$ then

$$\mathcal{R}(P) \subset \mathcal{R}(Q) \tag{2.1}$$

$$\Omega(P) \subset \Omega(Q). \tag{2.2}$$

2.1.3. *Examples and discussion.*

Example 2.1 ((Context free) simple Technology)**.** A *simple technology* is a technology in which the underlying activities are of the following form

(1) $\sigma_i \to \sigma_j$

where σ_i and σ_j are distinct descriptors. This is a very simple technology because all input states stay of the same length with the elements of the string changing one at a time.

Do we need to add ability to write λ the empty string?

Example 2.2 (Regular Technology)**.** A *right regular technology* is a technology in which the underlying activities take one of the following forms.

[3]In general the order in which a fixed set of activities can be feasibly applied to a given state in order to yield the same outcome is not unique. For example, consider $1 + 2 + 5$. The routine here would be $\rho : ++$, however the partial sums can be calculated from the left first, i.e. $1 + 2 = 3$, then $3 + 5 = 8$ or from the right $2 + 5 = 7$ then $1 + 7 = 8$. To avoid this kind of trivial multiplicity we will typically assume that all activities to be applied at the left of a string are fully resolved before moving to the next element to the right in the string, and so on. This process of moving from left to right is called *left derivation*.

(1) $\sigma_i \to \sigma_j$
(2) $\sigma_i \to \sigma_i\sigma_j$
(3) $\sigma_i \to \lambda$

where σ_i and σ_j are distinct descriptors and λ is the empty string.

In a right regular technology productions states are only generated by switching individual descriptors or by adding single descriptors at the right hand end of a string (or by producing the null state/an empty string).

In an equivalent way one can define a left regular technology.

Example 2.3 ((Context sensitive) right simple Technology)**.** Let S_1 and S_2 be two disjoint sets of descriptors. We call S_1 input descriptors and S_2 output descriptors. A string of only input descriptors is an input state and and string of only output descriptors is an output state.

A *simple technology* is a technology in which the underlying activities are of the following form

(1) $s_2\sigma_i s_1 \to s_2\sigma_j s_1$

where $\sigma_i \in S_1$ and $\sigma_j \in S_2$ are individual descriptors, and s_1 and s_2 are finite (possibly the empty string λ strings from S_1^* and S_2^* respectively . This is a like a simple technology except the rewriting must occur in order from the left to the right, one descriptor at a time. Note of course the total length of the input, output and intermediate strings are of equal length.

View of technology as a collection of elementary production activities is implicit in literature. People, in fact, have often thought this way without formally stating it.

Examples: artisans, pin factory, assembly lines.

Even creative work can be though about in this way: study creative writing, enumerate the plots, etc.

Example of how hard to find a good routine: central line checklist from Checklist Manifest. Central lines must have been around for years and then the routine dropped infection rate.

Bureaucratic government manuals for how to follow procedures or writing.

Military field manual.

These are all examples of codified activities, but of course in many settings in which activities are used they are tacit knowledge.

3. Expanding Technology

The technology of a firm could expand in a variety of ways: accidental discovery, R&D or through acquisition. Acquisition might be purchasing or merging with another entity or more focused by licensing a patented technology.

Whatever the process which expands some a technology it will incur some cost and imply some benefit. The ways in which markets price technology transfer is not the focus of this paper, instead we focus on understanding the benefits of expanding a technology.

For concreteness we will often use the language of a merger to describe expanding technology but the same fundamental observations apply to any initial technology which is expanded by adding more know-how in the form of new technology.

3.0.1. *New ways for old things.*

Proposition 3.1. *Let A_1 and A_2 be two activity sets. Let $\mathcal{R}_M = \mathcal{R}(A_1 \cup A_2)$ be the technology produced from the union of the two activity sets, then*

$$\mathcal{R}(A_1) \cup \mathcal{R}(A_2) \subseteq \mathcal{R}_M$$

and there exist activity sets A_1 and A_2 such that

$$\mathcal{R}(A_1) \cup \mathcal{R}(A_2) \subsetneq \mathcal{R}_M$$

Proof. Consider some $s \in \Omega(A_1) \cap \Omega(A_2)$ such that s is produced by $\rho_1 \in \mathcal{R}(A_1)$ and $\rho_2 \in \mathcal{R}(A_2)$ with $\rho_1 \neq \rho_2$. If the routines ρ_1 and ρ_2 involve some common intermediate state s^* such that the two routines differ both on the transformation $\lambda \to s^*$ and on transformation $s^* \to s$ then there exist at least two new routines in $\mathcal{R}(A_1 \cup A_2) \notin \mathcal{R}(A_1) \cup \mathcal{R}(A_2)$. □

Except in some very special cases where there are no shared states or when the reachability and generative-ability of shared states are equivalent, merging technologies will lead to new routines for achieving the exisiting products of the firm. New routines are useful for two reasons, one static and one dynamic.

In a static world, with fixed prices, different routines will tend to lead to cost savings. For example if each of the firms was able to produce s at cost c but used different routines to do so, then if there is a shared state in the two routines there must be a strictly cheaper way to produce the output if that state has a different cost under the two routines.

Corollary 3.2. *If s can be produced by each of the individual firms at the same cost, say by the distinct routines ρ_1 and ρ_2, and there is some common intermediate state in the two routines which is produced with different cost then the merged firm can produce s at strictly lower cost than either of the individual firms.*

Proof. If total cost of s is the same and cost of intermediate state s' is different then chose cheapest way to produce s', say from firm i and use $\rho_{-i}(s', s)$ for the second half because it must be cheaper than $\rho_{-i}(s', s)$. □

The dynamic advantage of more routines is the option value of the flexibility they provide. If a firm has only one way of producing an output then it is unable to adjust to a change in an input price and hence its cost of producing the given output. A prerequisite for substituting to save costs is to have an alternative production routine.

3.0.2. *New things.* The real power of this framework is in showing how expanded knowledge of activities can lead to new products for a merged firm.

Proposition 3.3. *Let A_1 and A_2 be the technologies of firms 1 and 2 respectively. Then $\Omega(A_i)$ are the production sets for firm $i = 1, 2$ separately and we denote $\Omega_M = \Omega(A_1 \cup A_2)$ be the production set for the merger of these two firms, then*

$$\Omega(A_1) \cup \Omega(A_2) \subseteq \Omega_M$$

and there exist technologies A_1 and A_2 such that

$$\Omega(A_1) \cup \Omega(A_2) \subsetneq \Omega_M$$

.

Proof. jb □

Need a good example here. One of buying a firm to buy a set of new activities that fits well with what a firm already has.

Possible tech examples: look at acquisitions by google, apple, etc. Did someone buy voice recognition for Alex?

"Google announced that it has entered into an agreement to acquire Nest Labs, the maker of high-tech thermostats and smoke detectors, for $3.2 billion. The deal is the second biggest in Google's history after the $12.5 billion acquisition of mobile phone manufacturer Motorola. Through this acquisition Google will gain a firm footing in the growing market for web-connected household appliances. Moreover, this deal can also strengthen Google's footprint in an urban household. In this note, we will briefly look at Google's strategy behind this acquisition." from Jan 17, 2014, 02:57pm. Google's Strategy Behind The $3.2 Billion Acquisition Of Nest Labs

Dropbox bought firm for dropbox paper.

3.1. **Value of expanding a firm's technology.** Consider a firm acquiring another firm for their technology, which it knows. This produces new products but what are the value of the new products?

In general one can have an arbitrary function which assigns values to states in Ω, how for apploed modeling purposes it is useful to have a valuation function described by a simple rule, below we give some examples of such rules and apply them to the situation of synergies in mergers and acquisitions.

Definition 9. Let $\#(s, \omega)$ be the *string count* of the number of distinct occurrences of the string s in the output ω.

Some of the simplest, non-trivial, valuation functions are

Definition 10. A valuation function is *string linear* if for a price vector p we have

$$v(\omega) = p \cdot (\#(s_1, \omega), ..., \#(s_n, \omega)) \tag{3.1}$$

4. Reliability of Production: Routines

In order for the production described by a routine to be feasible in the all of the activities involved need to remain feasible. In this real world a variety of shock may either temporarily or permanently make individual activities infeasible affecting the technology and potentially the output set generated by the technology. Shocks might come from a number of real world causes:

(1) unavailable inputs to a particular activity due to natural disasters, wars, political factors etc;
(2) supply price shocks where an input to an activity increase to such a degree that using the associated activity is not longer economically practical given the cost of the activity relative to the value generated;
(3) regulation may prohibit the use of an activity, e.g. directly through animal welfare regulation prohibiting activities like battery chickens, or indirectly due to regulations prohibiting pollution or toxic inputs like lead paints.

One natural way to analyse the potential impact of the loss of activities on a technology is through the concept of *reliability*. Reliability measures capture how susceptible a technology is to loosing the feasibility of outputs due to the loss of activities. There are many such reliability measures, we define some key types below and discuss the relevant economic properties of reliability measures.

Besides being an important practical characteristic of a technology in its own right, reliability also provides another lens through which to consider the benefit of expanding the the number of activities on which a technology is based. The main idea we investigate is planned redundancy, which activity or sub-routine to add if it is possible to add an activity or a sub-routine (i.e. which has the greatest benefit).

In general statements on the benefits of planned redundancy are technology specific, however we can identify the reliability benefits of certain components of a technology, these help think locally about the benefits of certain building blocks in improving reliability.

5. Distributed Technology

Another advantage of breaking the overall mapping of inputs to outputs into the smaller know-how based components of our technology is that we can how describe how individuals working in an organisation might know only part of the technology known by the organisation.

There are many applications from traditional organisational economics approaches that can be described as private information about the technology of a firm (Aghion and Tirole, 1997), or costly knowledge (Garicano, 2000; Garicano and Rossi-Hansberg, 2006). One characteristic of these familiar models is that the activities of the agents is production are unrelated. In this section we use our previous reliability framework to consider issues of distributed private knowledge under interdependent production activities.[4]

Definition 11. The *knowledge of individual* j about technology $\mathcal{R}(A)$, referred to as j's knowledge or technology, is $\mathcal{R}(A^j) \subset A$.

Individual j can only perform those routines that she know, i.e. are elements of her technology. This means on her own she is only able to perform a routine ρ if $\rho \in \mathcal{R}(A^j)$. However, a routine can be performed by a group of people, with different people performing different activities in the sequence given by ρ, as long as each person knows the activities they are to perform.

Definition 12. An *implementation* of a non-empty routine ρ on group G of K people is a function:

$$\mu : \{ \rho[1], \rho[2], \ldots, \rho[n] \} \to G^K \ ,$$

that assigns to every activity that comprises ρ to a responsible adult from G.

An implementation of a routine on a group assigns a person for each activity. A person can perform more than one activity or the same activity at different points in the routine. However it order for the a person to actually perform the activity they must know the activity (it must be in their knowledge set or technology), this is captured for the implementation by the following definition of feasibility.

Definition 13. An implementation of a routine ρ of length $n \geq 1$ is *feasible* if each $i \in \{1, ..., n\}$ it holds that $\rho[i] \in A^{\mu(\rho[i])}$.

We now investigate how, in a distributed technology, the inter-related choice of individual knowledge and implementation affect the reliability of a

[4]The information processing model of Radner (1992), Van Zandt and Radner (2001),Van Zandt (1999) have interdependent tasks and the idea of reliability in decision making features in Meagher (1996)

routine. Focusing on the role of workers, the most natural source of fragility is the absence of a worker who is supposed to be working in the implementation of a given routine. Workers are frequently absent due to illness or no longer being emplyed. We do not differentiate between these situation, focusing only on the immediate consequence of their absence.

If a worker is absent and no one can replace them from the group of available workers then .... which ever model we are using of reliability.

The solution to this loss is to have redundancy in the system so that another worker can step in to perform the missing activity. This requires that the alternative worker have the activity in their technology and is available to do it.

Availability: in a one shot model this is not an issue since the activities occur sequentially. This becomes a problem in a through put model if a worker needs to perform an activity in a later implementation of a routine.

Feasibility. Maximum resilience for a routine is achieved by having as large a number of potential replacements as possible, each with full knowledge of all the activities in the routine. In practice this would be prohibitively expensive so we now introduce the cost of knowledge in to our analysis of reliability.

Definition 14. The cost per period of a worker i is an increasing function of the number of activities they know are denoted $w(|A^i|)$ with $w' > 0$.

Unless otherwise stated, for technical convenience we assume that each activity can be performed in one period.

Consistent with the traditional approach we begin our characterization of a firm[5] with its *production possibility set*. The production possibility set is the set consisting of all feasible output sets for a firm. Given an infinite amount of time a firm can obviously produce as many of each product, feasible under its technology, as it likes. Thus it is critical to relate output to a time frame, we call this the *planning period* and hence refer to a production possibility set for planning period T.[6]

Before moving to the full infinite period model we start with the simpler one-shot model.

[5]We will refer to any organization undertaking production with distributed technology as a firm, recognizing that production also occurs in the public sector and in not-for-profits.
[6]For the infinite period model the planning period will focus on steady state production, that is intermediate production states will be allowed as inputs if they are produced by at the end of each planing period. For example in looking at the output of a car production line per day in steady state we would not assume that all incomplete cars are thrown away at the end of each day and that all productions starts afresh, with no carry over, each morning.

5.1. **One Shot Model.** The one-shot framework deals with all the single outputs that can be produced from scratch in a given planning period. Thus it considers all variations of a single routine is performed exactly once. This is the simplest case and eliminates complicated issues of scheduling and throughput. There still remain important issues around organizational design and utilization of workers.

The organizational design problem is to choose a set of workers and their individual knowledge sets in order to be able to perform the desired routines. For any routine, the lowest average wage across workers is to hire/train workers so that they know only one activity. This corresponds to Taylorism and mass production.

However having multiple workers raises the questions of what kind of employment contract applies. Are workers hired on a spot market and paid only for the periods in which they work (which we will call a spot market) or are workers hired on a continuing basis, i.e. salaried, and paid for each period regardless of whether they are busy of not.

These two dimesions, knowledge and the employment contract are a parallel of the tradition craft organisation and industrial production as discussed in Stinchcombe (1990).

Proposition 5.1. *For a simple technology or context sensitve simple technology the optimal organisation involves,* ***check this prop for production sets rather than random problems***

(1) under spot employment it is always cheaper to have n workers who each know only one distinct thing.
(2) under salaried employment if is cheaper to have one employee who knows the 2 activities as opposed to 2 workers who each know 1 thing, if and only if $w(2) < 2w(1)$.
(3) for length n strings and $k > 2$ activities then for each s there exist functions f_s such that it is optimal to hire a pool of workers such that each knows s.

Insert industrial revolution example.

The Taylorism solution, where each activity is know by only one person is cheap buy very fragile. In anyone is absent the whole routine stops. The craft organisation is also fragile since there is only one person implying that their absence also stops production.

Under a spot contract a worker is only paid for the work they do. Under the salary regime works get paid for the total duration of time it takes to execute the implementation of the routine. Obviously a firm would prefer the former regime but external circumstances may impose the latter. For

example, when the routine occurs at a remote location like a mine, an offshore oil rig or a film set in the dessert crew may need to be on location and employment law may stipulate this is for the duration of the project. See Meagher and Van Zandt (1998); de Fontenay and Meagher (2001)

5.1.1. *Digression: reliability in the one-shot.* Modern production, especially Lean Manufacturing, favour more resilient arrangements emphasizing training and job rotation. As a digression, we can apply the ideas on reliability from earlier to the one shot problem.

Assume the cost of an implementation of routine ρ, denoted $c(\rho, \mu(\rho))$ is given by

$$c(\rho, \mu(\rho)) = pF(\rho, \mu(\rho), \bar{A}) + \sum_{i=1}^{n} w(A^{\mu(\rho[i])}) \tag{5.1}$$

where $\bar{A} = \{A^1, ..., A^n\}$ and p is the cost to the firm of fragility.

Proposition 5.2. *If emplyment is on a spot contract and under* ***conditions*** *it is optimal to have workers know more than one thing. Condition for each to know 2...*

6. An organizational model of production with distributed know how

We propose a series of models with decreasing omniscience on the part of the designers.

(1) Task is only string rewriting
   (a) know whom to pass to for the next step v not knowing.
   (b) people knowing more than one thing.
   (c) 'bouncing' — workflow can go away and come back. e.g. output abba and 2 people know a and b each. flow of product is 1st person, 2nd person 2 periods and back to first person.
(2) introducing a new class 'manager' who don't know underlying prductive routines but know how to put (activity,person) pairs in to a process to solve a problem/be productive on an input. They are producing a finite state automata which is the plan/process for producing the outcome from an input. Take an imputable way first, i.e. plan at the beginning to solve all inputs that choose to solve.
(3) extensions: unforeseeen states, rare unprofitable states (might choose not to do them), structural breaks/ time evolving probabilities rare events becomes common for a while. These are all cases where a manager needs (potentially) to adjust the plans/organisation in real time.

6.1. **The Model.** Assume time is indexed by the integers $t = 1, 2, \ldots$. One input string arrives every period. The task of the organisation is to process these strings to produce outputs according to the organisations technology.

Let Σ be the set of descriptors and Σ^* the set of production states.

As above, there are a set G of workers indexed $j = 1, \ldots, K$ and each worker knows activities described by the set A^j. The cost of a worker depends on their knowledge set, i.e. $w(|A^j|)$ where $w' > 0$. We start with the simplifying assumption that only the total number of activities matters for cost, not which activities are known. We normalise the times it takes for each activity to one unit of time.

We characterise the performance of an organisation in the traditional way through its production possibility set. Planning horizon

The organisational design problem requires choosing a set of workers and their associated knowledge stock in order to be able to sustain some infinite production plan. Issues are both delay on each individual input and throughput, as well as efficiency with regard to labour inputs.

6.1.1. *Digression: Production function models.* Most traditional models of production within organisations consist of production processes which have routines of length one, that is in one step the 'black box' of a firm maps an input to an output if that output is feasible, or produces nothing if no output is feasible. Standard principal-agent models have an agent choosing in one time period a single action. The Garicano model has the structure ... Grossman and Hart, Aghion and Tirole ....

We begin by showing that if the single actions of traditional models are actually routines, made of potentially multiple activities, then the problems of organisation are significantly more subtle.

To illustrate the issue consider a simple technology where the input and output sets are both of size L.

Example 6.1 (l descriptor strings)**.** Consider a simple context free technology. Assume that the set of input strings are ordered pairs from $\{a_1, a_2, ..., a_L\}^2$ and the set of outputs is ordered pairs from $\{A_1, A_2, ..., A_L\}^2$ and the activities are of the form $a_i \to A_i$ for $i = 1, ..., L$.

In the traditional single step conception of production each of the input pair is matched by a seperate action to a unique output, thus there must be L^2 different actions, one for each routine. However, from the know-how perspective and given this is a simple technology there are in fact only L pieces of knowledge to know.

Proposition 6.2. *Something like: For a simple technology with $d = \infty$ and w increases sufficiently fast the single step model oversates the cost of*

knowledge by $\mathcal{O}(L^2)$ and has as the optimal solution as $2L$ workers with wages of $w(L^2)$ where as the true optimal solution is $2L$ workers with wages of $w(L)$.

Proof. Check this theorem is true. number of ordered pairs is L^2, but there are really only L activities. □

Show (if) the output set has standard neoclassical properties - there aren't many.

6.2. **Results.**

Example 6.3 (2 descriptor strings)**.** Assume that the set of input strings ordered pairs from $\{a, b\}^2$ and the set of output is ordered pairs from $\{A, B\}^2$ in which each lower case letter is rewritten to its corresponding upper case. Two cannocial versions of this problem are context free and the context sensitive.

Further let us assume the all activities are *simple*

Now we are doing production possibility sets we will focus on regular technologies

We have some lemmas on characterizing workflows for sets of outputs based on the number of each letter in total in the outputs.

By example even with regular (linear) technology the optimal workflow can be cyclic eg profitable to have 1 know a,c and 2 know b and then be writing abbc. Work flow will be 1,2,2,1 to writ a, ab, abb, abbc.

This also suggests another general proposition something like If it is profitable/required to have someone know more than one thing then e,g 2 person case and a,b,c, the person who knows 2 things will know the 2 least frequent things. Insight kind of knapsack problem and wantng to balance out the workloads.

7. Conclusion

We have developed a new framework for technology based on the idea of know how as small activities which can be applied in a variety of situations.

The framework is very powerful yielding immediately the possibility of new products and cost synergies from mergers which are impossible under the Neoclassical approach to technology.

We also show how our approach provides a consistent framework for thinking about the organisation of work inside organisations, especially when knowledge is dispersed. Many of the exisiting models appear as special cases of our framework and the extra level of generality makes clear the special assumptions within those individual models.

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